

■ B.TECH FIRST YEAR · ENGINEERING CHEMISTRY

Unit III: Electrochemistry & Applications

A Complete Zero-to-Expert Study Module

📖 Subject: Engineering Chemistry

🎓 Level: B.Tech 1st Year

📄 20+ Page Comprehensive Module

🔬 Unit III – Electrochemistry

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 **How to Use This Module:** Every concept is explained from zero level. Read each section sequentially. Pay special attention to formula boxes, step-by-step solutions, and SVG diagrams. Use the Quick Revision Sheet before exams.

§1 Introduction to Electrochemistry

✓ STEP 1 – BASIC IDEA

Electrochemistry is the branch of chemistry that deals with the relationship between **electrical energy and chemical reactions**. In simple terms: some chemical reactions can *produce* electricity, and electricity can *cause* chemical reactions.

Real-Life Analogy: *Think of a torch battery. Inside it, chemicals react and produce electricity that lights the bulb. That's electrochemistry in action! Conversely, when you charge your phone, electricity is used to reverse a chemical reaction — also electrochemistry.*

✓ STEP 2 – WHY IT MATTERS IN ENGINEERING

- Design and develop **batteries** for electric vehicles, smartphones, and laptops
- Manufacture **fuel cells** for clean energy generation
- Build **electrochemical sensors** for medical devices (glucose monitors)
- Understand **corrosion** and prevent metal degradation
- Perform electroplating, electrorefining in metallurgy

✓ STEP 3 – FUNDAMENTAL CONCEPT: OXIDATION-REDUCTION

All electrochemical processes involve **redox reactions**:

- **Oxidation:** Loss of electrons → occurs at the *Anode*
- **Reduction:** Gain of electrons → occurs at the *Cathode*

Memory trick: **OIL RIG** → Oxidation Is Loss, Reduction Is Gain

⚡ KEY ELECTROCHEMICAL RELATIONSHIPS

Charge (Q) = Current (I) × Time (t) [Q in Coulombs, I in Amperes, t in seconds]

1 Faraday (F) = 96,485 C/mol ≈ 96,500 C/mol

Faraday = charge of 1 mole of electrons

$$\text{Electrical Energy} = Q \times E \quad [\text{Joules} = \text{Coulombs} \times \text{Volts}]$$

🔑 Key Points:

- Electrochemistry connects chemistry and electrical engineering
- Redox reactions are the heart of every electrochemical process
- Faraday's constant (96,500 C/mol) is essential for all calculations
- Anode = Oxidation; Cathode = Reduction (Always remember this!)

§2 Fundamentals of Electrochemical Cells

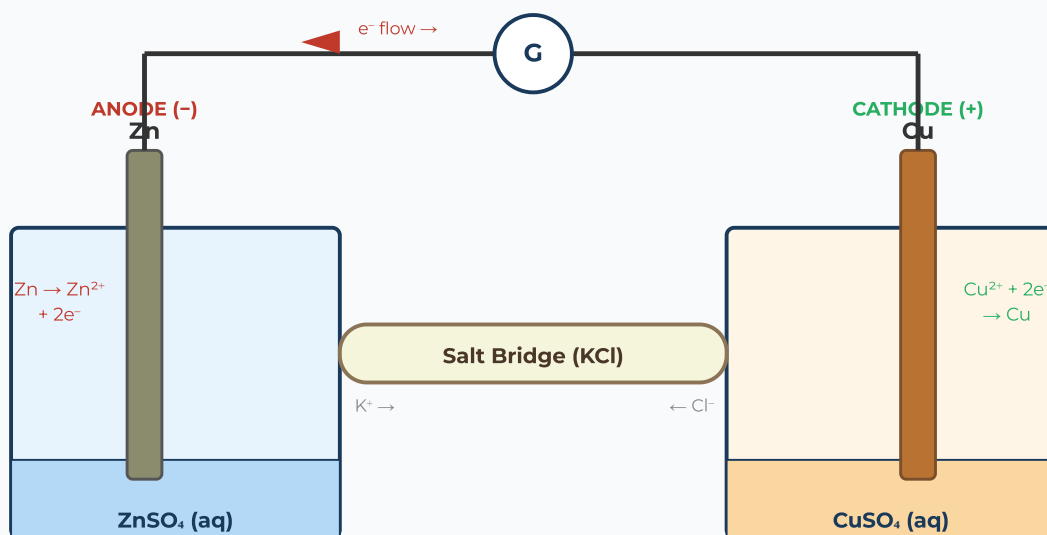
✅ STEP 1 – WHAT IS AN ELECTROCHEMICAL CELL?

An **electrochemical cell** is a device that converts chemical energy into electrical energy (or vice versa) using oxidation-reduction reactions. It consists of two **electrodes** (conductors) dipped in an **electrolyte** (ion-conducting solution).

Analogy: Imagine two buckets of water connected by a pipe. One bucket loses water (oxidation) and the other gains it (reduction). Electrons flow through a wire like water through the pipe — creating a current!

2.1 Types of Electrochemical Cells

Feature	Galvanic Cell (Voltaic Cell)	Electrolytic Cell
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Reaction	Spontaneous ($\Delta G < 0$)	Non-spontaneous ($\Delta G > 0$)
Power Source	Self-powered (generates EMF)	Needs external power supply
Anode Polarity	Negative (-)	Positive (+)
Cathode Polarity	Positive (+)	Negative (-)
Examples	Daniel cell, batteries, fuel cells	Electroplating, electrolysis of water
Salt Bridge	Required (usually)	Not needed (single electrolyte)



Daniell Cell (Galvanic Cell)

Fig. 2.1 – Daniell Cell showing Zinc anode, Copper cathode, Salt Bridge, and Electron Flow

2.2 Cell Notation (Representation)

✓ HOW TO WRITE CELL NOTATION

A cell is represented using a standardized notation. The rules are:

- **Anode** (oxidation) is written on the **LEFT**
- **Cathode** (reduction) is written on the **RIGHT**
- A **single line |** represents a phase boundary (between electrode and electrolyte)
- A **double line ||** represents the salt bridge

CELL NOTATION FORMAT

Anode | Anode electrolyte || Cathode electrolyte | Cathode

Example (Daniell Cell): Zn | ZnSO₄(aq) || CuSO₄(aq) | Cu

Meaning: Zn is oxidized (left) → Cu²⁺ is reduced (right)

2.3 Role of Salt Bridge

The salt bridge is a U-shaped tube filled with concentrated electrolyte (like KCl or KNO₃ in agar gel). Its functions are:

- Maintains **electrical neutrality** in both half-cells
- Completes the **electrical circuit** through ion migration

- Prevents **direct mixing** of the two solutions
- Allows ions to flow: K^+ migrates to cathode compartment, Cl^- migrates to anode compartment

§3 EMF and Cell Potential

✓ STEP 1 – WHAT IS EMF?

The **Electromotive Force (EMF)** is the maximum potential difference between the two electrodes of a cell when *no current is flowing*. Think of it as the "electrical pressure" that drives electrons from anode to cathode.

Analogy: EMF is like the water pressure difference between a high tank and a low tank. The higher the pressure difference, the more forcefully water flows. Similarly, higher EMF means electrons flow with more "force" (energy).

EMF FORMULAS

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} \quad [\text{Standard Reduction Potentials}]$$

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{right}} - E^{\circ}_{\text{left}} \quad [\text{Right} = \text{Cathode}, \text{Left} = \text{Anode}]$$

$$\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}} \quad [n = \text{electrons transferred}, F = 96500 \text{ C/mol}]$$

If $E^{\circ}_{\text{cell}} > 0 \rightarrow$ Spontaneous reaction (Galvanic)

If $E^{\circ}_{\text{cell}} < 0 \rightarrow$ Non-spontaneous (Electrolytic)

3.1 Standard Electrode Potential (E°)

The **Standard Electrode Potential** is the potential of an electrode measured under standard conditions:

- Temperature: 298 K (25°C)
- Ion concentration: 1 M
- Gas pressure: 1 atm

The **Standard Hydrogen Electrode (SHE)** is assigned $E^{\circ} = 0.00 \text{ V}$ and is used as the reference.

3.2 Electrochemical Series (Selected Values)

Electrode Reaction (Reduction)	E° (V)	Nature
$\text{F}_2 + 2\text{e}^- \rightarrow 2\text{F}^-$	+2.87	Strong Oxidant
$\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$	+1.36	Strong Oxidant
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$	+1.23	Oxidant
$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	+0.34	Mild Oxidant
$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (SHE)	0.00	Reference
$\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$	-0.76	Mild Reductant
$\text{Li}^+ + \text{e}^- \rightarrow \text{Li}$	-3.04	Strong Reductant

Important Rule: In a galvanic cell, the electrode with HIGHER reduction potential acts as cathode (reduction). The electrode with LOWER reduction potential acts as anode (oxidation).

✓ EXAMPLE CALCULATION

Find E°_{cell} for the Daniell cell: $\text{Zn} \mid \text{ZnSO}_4 \parallel \text{CuSO}_4 \mid \text{Cu}$

$E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ (cathode), $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ (anode)

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 - (-0.76) = 0.34 + 0.76 = +1.10 \text{ V}$

Since $E^\circ_{\text{cell}} > 0$, the reaction is spontaneous. ✓

§4 Nernst Equation – Concept & Applications

✓ STEP 1 – WHY STANDARD CONDITIONS ARE NOT ALWAYS REAL

The standard EMF (E°) is measured at exactly 1 M concentration, 25°C, and 1 atm. But in *real life*, concentrations change as reactions proceed. The **Nernst Equation** corrects for these real conditions.

Analogy: Imagine you're pumping water uphill. Standard EMF tells you the force at the beginning. But as the bucket fills up (concentration changes), the force changes. The Nernst Equation tells you the **ACTUAL** force at any moment.

✓ STEP 2 – INTUITIVE DERIVATION OF NERNST EQUATION

Starting from thermodynamics: The Gibbs free energy change is:

$$\Delta G = \Delta G^\circ + RT \ln Q \quad (Q = \text{reaction quotient})$$

Since $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$, substituting:

$$-nFE = -nFE^\circ + RT \ln Q$$

Dividing by $-nF$:

$$E = E^\circ - (RT/nF) \ln Q$$

At 298 K, converting $\ln$ to $\log_{10}$: $RT/F = 0.02569 \text{ V}$, so $(2.303 \times RT/F) = 0.05916 \text{ V} \approx 0.0591 \text{ V}$

⚡ NERNST EQUATION

$$E = E^\circ - (RT/nF) \times \ln Q \quad [\text{General Form}]$$

$$E = E^\circ - (0.0591/n) \times \log Q \quad [\text{At 298 K, using } \log_{10}]$$

Where: E = cell potential at given conditions (V)

E° = standard cell potential (V)

$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ (Gas constant)

T = Temperature in Kelvin

n = number of electrons transferred

$$F = 96500 \text{ C mol}^{-1} \text{ (Faraday's constant)}$$

$$Q = [\text{Products}]/[\text{Reactants}] \text{ (Reaction Quotient)}$$

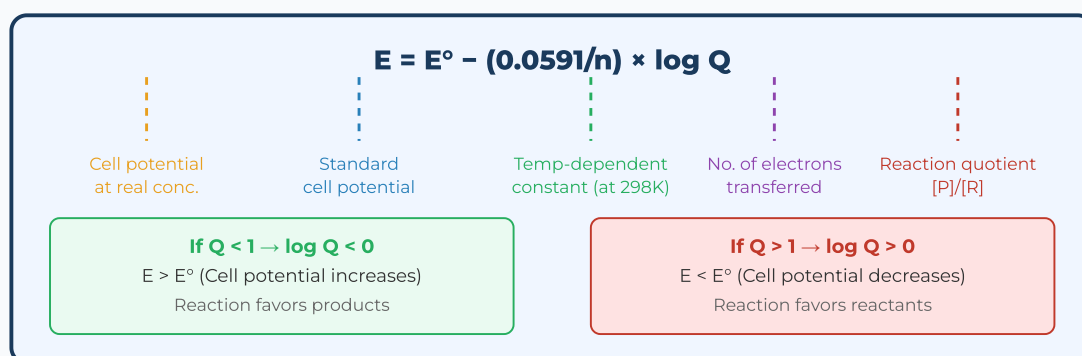


Fig. 4.1 – Nernst Equation: Components and Effect of Q on Cell Potential

4.1 Nernst Equation for Half-Cell

HALF-CELL NERNST EQUATION

For: $M^{n+} + ne^- \rightarrow M$

$$E = E^\circ - (0.0591/n) \times \log(1/[M^{n+}])$$

$$E = E^\circ + (0.0591/n) \times \log[M^{n+}]$$

At equilibrium: $E = 0$ and $Q = K_{eq}$

$$\text{Then: } E^\circ = (0.0591/n) \times \log K_{eq}$$

4.2 Applications of Nernst Equation

- Calculate actual cell EMF at non-standard concentrations
- Determine equilibrium constants (K) from standard potentials
- Understand pH measurement using glass electrode
- Study concentration cells (cells with same electrodes but different concentrations)
- Analyze corrosion potential in engineering materials

§5 Cell Potential Calculations

5.1 Concentration Cells

✓ WHAT IS A CONCENTRATION CELL?

A concentration cell is a galvanic cell where **both electrodes are made of the same material**, but they are in contact with electrolytes of **different concentrations**. The EMF is generated purely due to concentration difference.

CONCENTRATION CELL EMF

$E^{\circ}_{\text{cell}} = 0$ (same electrodes)

$E_{\text{cell}} = -(0.0591/n) \times \log([C_1]/[C_2]) = (0.0591/n) \times \log([C_2]/[C_1])$

Where $C_2 > C_1$ (cathode concentration > anode concentration)

5.2 Relationship between EMF, ΔG , and Equilibrium Constant

THERMODYNAMIC RELATIONSHIPS

$\Delta G = -nFE$ [G in Joules]

$\Delta G^{\circ} = -nFE^{\circ}$ [Standard conditions]

$\Delta G^{\circ} = -RT \ln K_{\text{eq}} = -2.303 RT \log K_{\text{eq}}$

Therefore: $E^{\circ} = (0.0591/n) \times \log K_{\text{eq}}$ [at 298 K]

$\log K_{\text{eq}} = nE^{\circ}/0.0591$

Summary of Signs:

E°_{cell}	ΔG°	K	Nature of Reaction
Positive (+)	Negative (-)	$K > 1$	Spontaneous (Galvanic)
Negative (-)	Positive (+)	$K < 1$	Non-spontaneous (Electrolytic)
Zero (0)	Zero (0)	$K = 1$	At Equilibrium

§6 Potentiometry – Basic Concepts

✓ STEP 1 – BASIC IDEA OF POTENTIOMETRY

Potentiometry is an analytical technique that measures the *potential (voltage)* of an electrochemical cell to determine the concentration of ions in a solution. It measures potential when NO current flows (or negligible current).

Analogy: *It's like reading a thermometer — you're measuring something without disturbing the system. In potentiometry, you measure voltage without drawing current, so the solution is undisturbed.*

✓ STEP 2 – BASIC PRINCIPLE

Based on the **Nernst equation**, the potential of an electrode depends on the concentration of ions. By measuring the voltage between an **indicator electrode** (sensitive to the ion of interest) and a **reference electrode** (constant potential), we can calculate ion concentration.

6.1 Components of a Potentiometric Setup

Component	Function	Example
Indicator Electrode	Responds to ion concentration (variable potential)	Glass electrode for H ⁺ , Ion-selective electrode
Reference Electrode	Provides constant, known potential	SCE (Saturated Calomel Electrode), Ag/AgCl electrode
High-impedance Voltmeter	Measures voltage without drawing current	pH meter, Potentiometer
Salt Bridge/Junction	Connects two electrode solutions	KCl solution

6.2 Reference Electrodes

STANDARD CALOMEL ELECTRODE (SCE)

Reaction: $\text{Hg}_2\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Hg} + 2\text{Cl}^-$

$E(\text{SCE}) = +0.242 \text{ V vs SHE (for saturated KCl)}$

Notation: $\text{Hg} \mid \text{Hg}_2\text{Cl}_2(\text{s}) \mid \text{KCl}(\text{sat'd}) \parallel \text{solution}$

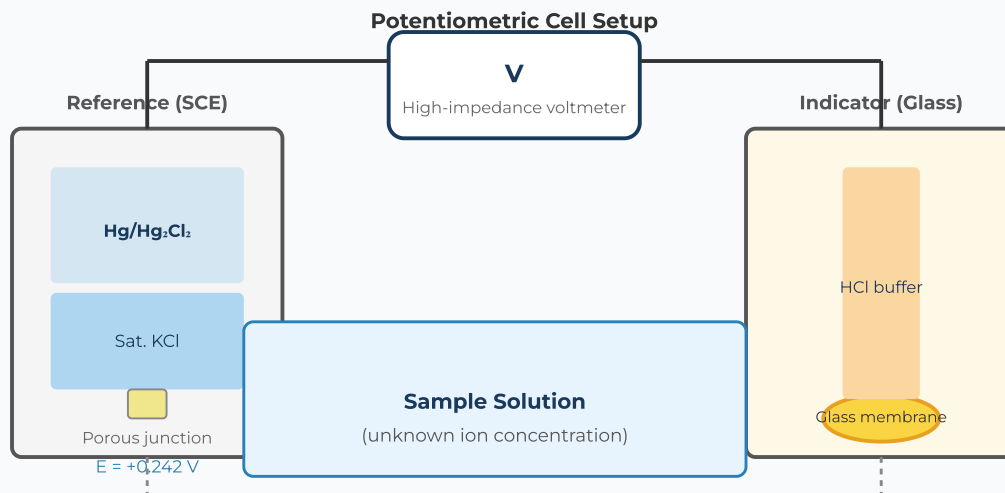


Fig. 6.1 – Potentiometric measurement setup with SCE (reference) and Glass electrode (indicator)

6.3 pH Measurement Using Glass Electrode

PH AND ELECTRODE POTENTIAL

$$E_{\text{glass}} = K - 0.0591 \times \text{pH} \quad [\text{at } 298 \text{ K, } K = \text{constant}]$$

$$E_{\text{cell}} (\text{measured}) = E_{\text{glass}} - E_{\text{reference}}$$

$$\text{pH} = (E^\circ - E_{\text{measured}}) / 0.0591$$

Sensitivity: 59.1 mV per pH unit change at 25°C

§7 Potentiometric Titrations (Redox)

✓ STEP 1 – WHAT IS POTENTIOMETRIC TITRATION?

In a **potentiometric titration**, instead of using an indicator (like phenolphthalein), we *measure the cell potential (E)* as the titrant is added. The **equivalence point** is detected by a sudden sharp change in potential.

Analogy: *Imagine walking in the dark and suddenly hitting a step — that sharp change in surface tells you where the step is. In potentiometric titration, the sudden jump in voltage tells you where the equivalence point is.*

✓ STEP 2 – HOW IT WORKS (REDOX TITRATION)

1. The sample (analyte) is placed in a beaker with an **indicator electrode** and a **reference electrode**.
2. The titrant is added in small increments from a burette.
3. After each addition, the cell potential (E) is recorded.
4. A graph of E vs. Volume of titrant is plotted.
5. The **equivalence point** corresponds to the steepest part (inflection point) of the curve.

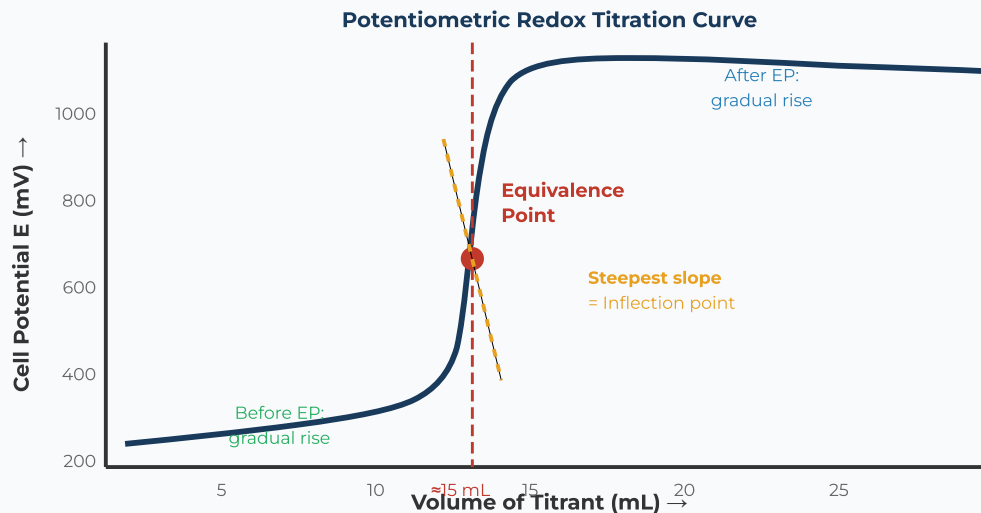


Fig. 7.1 – E vs. Volume curve in potentiometric titration. Sharp inflection = Equivalence Point.

7.1 Graph Interpretation – How to Find Equivalence Point

METHODS TO LOCATE EQUIVALENCE POINT

Method 1 – Inflection Point: Point of steepest slope on E vs V curve

Method 2 – First Derivative: Plot $\Delta E/\Delta V$ vs V $\rightarrow$ maximum = EP

Method 3 – Second Derivative: Plot $\Delta^2 E/\Delta V^2$ vs V $\rightarrow$ zero crossing = EP

7.2 Advantages of Potentiometric Titration

- No need for colored indicators — **applicable to colored or turbid solutions**
- More **precise and accurate** than visual indicator methods
- Can detect **multiple equivalence points** in complex mixtures
- Suitable for **acid-base, redox, precipitation, and complexometric** titrations
- Can be **automated** for industrial quality control

7.3 Common Redox Potentiometric Titrations

Titration	Analyte	Titrant	Indicator Electrode
Iron(II) with Permanganate	Fe ²⁺	KMnO ₄	Pt electrode
Dichromate titration	Fe ²⁺	K ₂ Cr ₂ O ₇	Pt electrode
Iodometric titration	Reducing agents	I ₂ in KI	Pt electrode
Cerium(IV) titration	Various reductants	Ce(SO ₄) ₂	Pt electrode

§8 Conductivity – Basic Concepts

✓ STEP 1 – WHAT IS ELECTRICAL CONDUCTIVITY?

Conductivity is the ability of a material to allow electric current to flow through it. For ionic solutions (electrolytes), conductivity depends on the **number of ions** and their **mobility** in solution.

Analogy: Think of a highway. More cars (ions) moving faster = more conductivity. A crowded highway where cars move slowly = lower conductivity. Adding more salt (electrolyte) = more cars on the road!

✓ STEP 2 – KEY DEFINITIONS

- **Resistance (R):** Opposition to current flow [Unit: Ohm, Ω]
- **Conductance (G):** $G = 1/R$ [Unit: Siemens, $S = \Omega^{-1}$]
- **Specific Conductance (κ):** Conductance of unit cube ($1\text{ cm} \times 1\text{ cm} \times 1\text{ cm}$) of solution
- **Molar Conductance (Λ_m):** Conductance of solution containing 1 mole of electrolyte between electrodes 1 cm apart
- **Equivalent Conductance (Λ_{eq}):** Conductance of 1 gram-equivalent of electrolyte

CONDUCTIVITY FORMULAS

Conductance: $G = 1/R$ [S or mho]

Specific Conductance: $\kappa = G \times (l/A) = (l/A) \times (1/R)$ [S/cm or S/m]

Cell Constant: $K_{\text{cell}} = l/A$ [cm^{-1}]

Therefore: $\kappa = G \times K_{\text{cell}}$

Molar Conductance: $\Lambda_m = (\kappa \times 1000) / C$ [C in mol/L, Λ_m in $\text{S} \cdot \text{cm}^2/\text{mol}$]

Equivalent Conductance: $\Lambda_{eq} = (\kappa \times 1000) / N$ [N in eq/L]

8.1 Factors Affecting Conductivity

Factor	Effect on Conductivity	Reason
Concentration $\uparrow$	κ increases then decreases	More ions, but ionic interaction increases at high conc.
Temperature $\uparrow$	κ increases	Ions move faster; viscosity decreases
Nature of electrolyte	Strong > Weak electrolytes	Degree of dissociation differs
Ionic charge	Higher charge $\rightarrow$ higher conductance	More charge carriers
Ionic mobility	Higher mobility $\rightarrow$ higher conductance	H^+ and OH^- have anomalously high mobility

8.2 Kohlrausch's Law

KOHLRAUSCH'S LAW OF INDEPENDENT MIGRATION OF IONS

$$\Lambda^{\circ m} = \lambda^{\circ+} + \lambda^{\circ-} \quad [\text{at infinite dilution}]$$

Where $\lambda^{\circ+}$ = molar conductance of cation at infinite dilution

$\lambda^{\circ-}$ = molar conductance of anion at infinite dilution

$$\text{For KCl: } \Lambda^{\circ m}(\text{KCl}) = \lambda^{\circ}(\text{K}^+) + \lambda^{\circ}(\text{Cl}^-) = 73.5 + 76.3 = 149.8 \text{ S}\cdot\text{cm}^2/\text{mol}$$

$$\text{Empirical equation: } \Lambda_m = \Lambda^{\circ m} - A\sqrt{C} \quad [A = \text{constant, } C = \text{concentration}]$$

§9 Conductivity Cell & Measurements

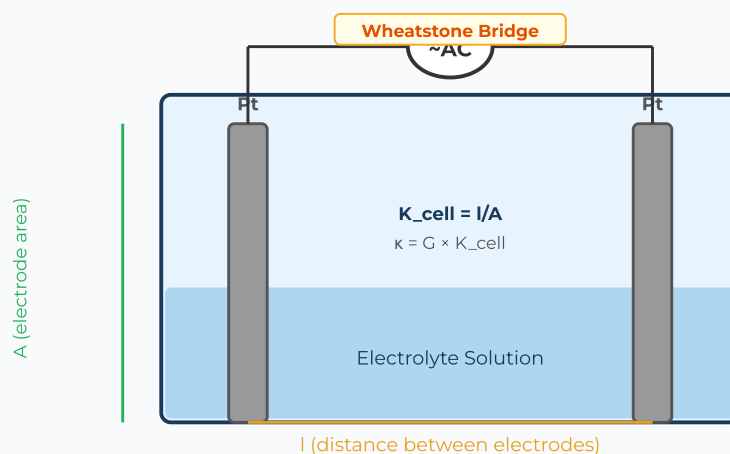


Fig. 9.1 – Conductivity Cell with platinized Pt electrodes, AC source, and Wheatstone bridge

9.1 Why AC (Alternating Current) is Used?

- AC prevents **electrolysis** of the solution (DC would decompose the electrolyte)
- AC prevents **polarization** of electrodes
- Frequency typically 1000–5000 Hz is used
- Conductance is measured using a **Wheatstone bridge** or digital conductometer

9.2 Platinization of Electrodes

The platinum electrodes are coated with **platinum black** (finely divided platinum) to:

- Increase the effective surface area
- Reduce electrode polarization
- Obtain accurate resistance measurements

§10 Conductometric Titrations

✓ STEP 1 – BASIC CONCEPT

In **conductometric titration**, the conductance of a solution is monitored as a titrant is added. Since different ions have different conductances, the addition of titrant changes the overall conductance in a predictable way. The **equivalence point** is located at the minimum (or kink) of the conductance curve.

Analogy: Imagine replacing fast runners (H^+ ions) with slow walkers (Na^+ ions) in a race. The total team speed (conductance) changes with each replacement. The minimum speed tells you when all the runners have been replaced!

10.1 Acid-Base Conductometric Titration: Strong Acid vs Strong Base

Example: HCl titrated with NaOH

Initial solution: $HCl \rightarrow H^+ + Cl^-$. H^+ has very high ionic mobility ($\lambda = 350 \text{ S}\cdot\text{cm}^2/\text{mol}$)

1. **Before EP:** As NaOH is added, $H^+ + OH^- \rightarrow H_2O$. Fast H^+ ions are removed and replaced by slower Na^+ ions → **conductance DECREASES**
2. **At EP:** All H^+ consumed. Solution has only NaCl. **Minimum conductance.**
3. **After EP:** Excess NaOH adds Na^+ and OH^- (OH^- also has high mobility) → **conductance INCREASES sharply**

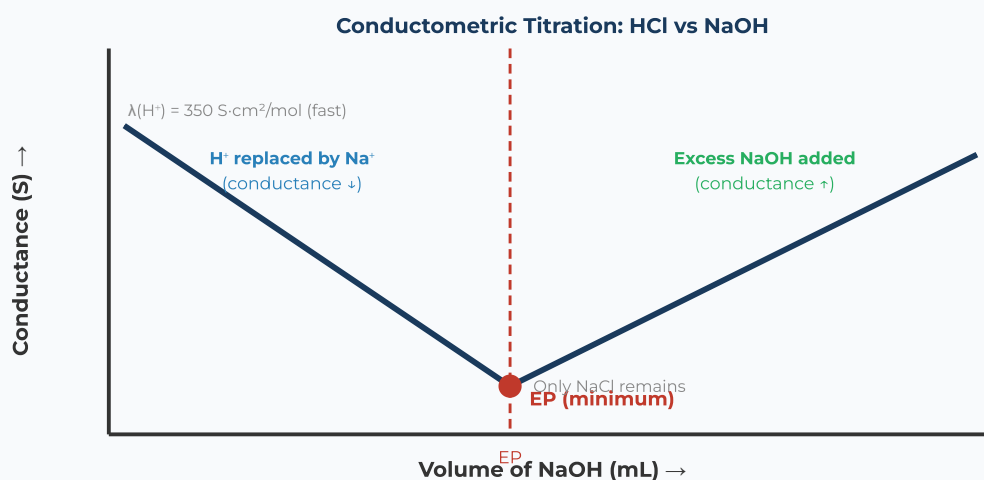


Fig. 10.1 – Conductometric curve: V-shaped for strong acid vs. strong base titration

10.2 Other Conductometric Titration Curves

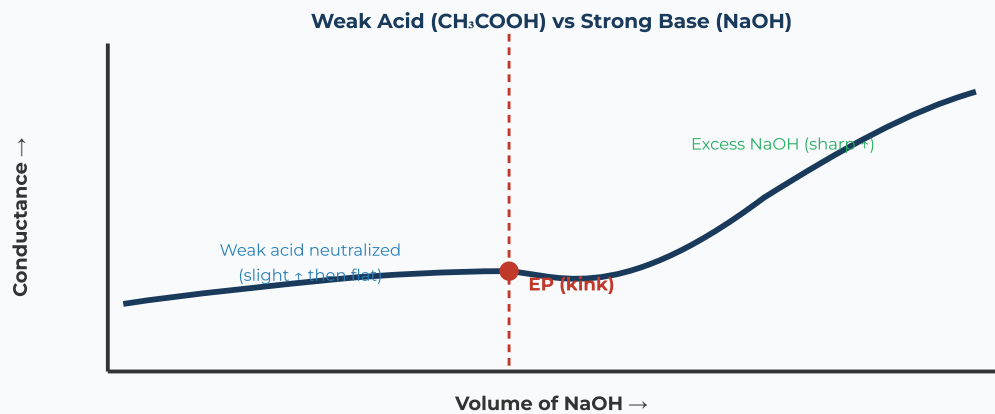


Fig. 10.2 – Conductometric curve for weak acid vs. strong base: gradual rise then sharp rise at EP

10.3 Comparison: Potentiometric vs Conductometric Titration

Feature	Potentiometric	Conductometric
Measured parameter	Cell potential (E, mV)	Conductance (G, S)
Indicator electrode	Required (ion-selective)	Not required (Pt electrodes only)
EP detection	Sharp E jump (inflection)	Minimum or kink in G curve
Dilute solutions	Works well	Less accurate (small G change)
Colored solutions	Suitable	Suitable
Complexity	More complex setup	Simpler setup
Applications	Redox, pH, pM titrations	Acid-base, precipitation

§11 Electrochemical Sensors

✓ BASIC CONCEPT

An **electrochemical sensor** is a device that detects the presence or concentration of a chemical species by converting a chemical reaction (at an electrode) into a measurable electrical signal. They are widely used in medical diagnostics, environmental monitoring, and industrial process control.

Types of Electrochemical Sensors

Type	What it Measures	Signal	Example
Potentiometric	Ion activity/concentration	Potential (voltage)	pH electrode, ISE
Amperometric	Analyte concentration	Current	Glucose sensor, O ₂ sensor
Conductometric	Ion concentration	Conductance	Salinity sensor
Voltammetric	Analyte identity & concentration	Current vs Voltage	Heavy metal detection

§12 Potentiometric Sensors

✓ STEP 1 – HOW POTENTIOMETRIC SENSORS WORK

Potentiometric sensors measure the **open-circuit potential** (no current flows) between an ion-selective electrode and a reference electrode. The potential changes according to the **Nernst equation**, directly proportional to the logarithm of ion activity.

NERNST RESPONSE OF POTENTIOMETRIC SENSOR

$$E = \text{const.} + (2.303RT/zF) \times \log[a_i]$$

$$\text{At } 298 \text{ K: } E = \text{const.} + (0.0591/z) \times \log[a_i]$$

Where: z = charge of ion, a_i = activity of ion

For H^+ ($z=1$): 59.1 mV per decade change in $[\text{H}^+]$

For Ca^{2+} ($z=2$): 29.6 mV per decade change in $[\text{Ca}^{2+}]$

12.1 Ion-Selective Electrodes (ISE)

ISE Type	Target Ion	Membrane Material	Application
Glass Electrode	H^+ (pH)	Special glass (SiO_2 lattice)	pH meters, food quality
Fluoride ISE	F^-	LaF_3 crystal	Water fluoride monitoring
Calcium ISE	Ca^{2+}	Liquid ion exchanger	Blood calcium analysis
Sodium ISE	Na^+	Glass membrane	Clinical chemistry
Nitrate ISE	NO_3^-	Quaternary ammonium salt	Water quality testing

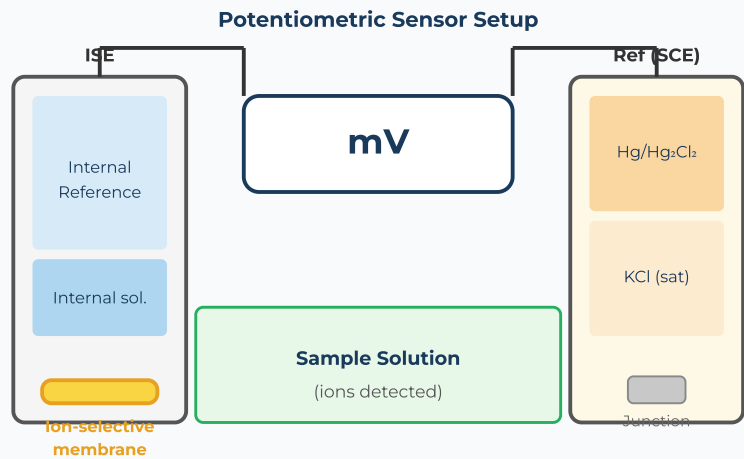


Fig. 12.1 – Potentiometric sensor with ISE and reference electrode measuring ion activity in sample

§13 Amperometric Sensors

✓ HOW AMPEROMETRIC SENSORS WORK

Amperometric sensors measure the **current** generated when an analyte is electrochemically oxidized or reduced at an electrode held at a fixed potential. The current is directly proportional to the concentration of the analyte (**Faraday's law**).

AMPEROMETRIC SENSOR PRINCIPLES

$i = nFADC/\delta$ (Limiting current equation)

Where: i = current (A), n = electrons transferred, F = 96500 C/mol

A = electrode area (cm^2), D = diffusion coefficient (cm^2/s)

C = analyte concentration (mol/cm^3), δ = diffusion layer thickness (cm)

Key: Current $\propto$ Concentration of analyte

13.1 Clark Oxygen Electrode (Amperometric)

The most famous amperometric sensor — the **Clark electrode** — measures dissolved oxygen in blood and water:

- A Pt cathode and Ag/AgCl reference anode are enclosed in a membrane-covered cell
- Applied potential: -0.6 V vs reference
- Reaction at cathode: $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
- Current generated is proportional to O_2 concentration

13.2 Glucose Amperometric Biosensor

Used in diabetic glucose meters:

- $\text{Glucose} + \text{O}_2 \rightarrow \text{Gluconic acid} + \text{H}_2\text{O}_2$ (catalyzed by glucose oxidase enzyme)
- $\text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{O}_2 + 2\text{e}^-$ (oxidized at Pt anode)
- Current measured $\propto$ Glucose concentration
- Reading displayed in mg/dL or mmol/L

Feature	Potentiometric Sensor	Amperometric Sensor
Measured signal	Voltage (E)	Current (i)
Principle	Nernst equation (log conc.)	Faraday's law (linear conc.)
Applied potential	None (open circuit)	Fixed external potential
Sensitivity	Logarithmic response	Linear response (better)
Examples	pH meter, ISE, F^- sensor	Glucose meter, O_2 sensor
Interference	Only from same-charge ions	Electroactive interferents

§14 Batteries – Primary & Secondary Cells

✓ STEP 1 – WHAT IS A BATTERY?

A **battery** is a collection of one or more electrochemical cells connected together to provide electrical energy. It converts stored chemical energy into electrical energy on demand through spontaneous redox reactions.

Analogy: A battery is like a pre-filled water tank. The water (chemical energy) flows out (as electricity) and does work. A secondary battery is like a tank you can refill — rechargeable!

Feature	Primary Battery	Secondary Battery
Rechargeability	Non-rechargeable (single use)	Rechargeable (multiple cycles)
Reaction	Irreversible	Reversible
Cost	Cheaper initially	More expensive but economical long-term
Energy density	Generally higher	Improving rapidly
Examples	Zinc-air, Leclanché cell	Li-ion, Lead-acid, NiMH
Applications	Hearing aids, watches, remotes	EVs, smartphones, laptops

14.1 Zinc-Air Battery (Primary)

✓ WORKING PRINCIPLE

The zinc-air battery uses **atmospheric oxygen** as the positive electrode (cathode) material. This makes it very light and energy-dense. The oxygen is consumed from air through a porous carbon cathode.

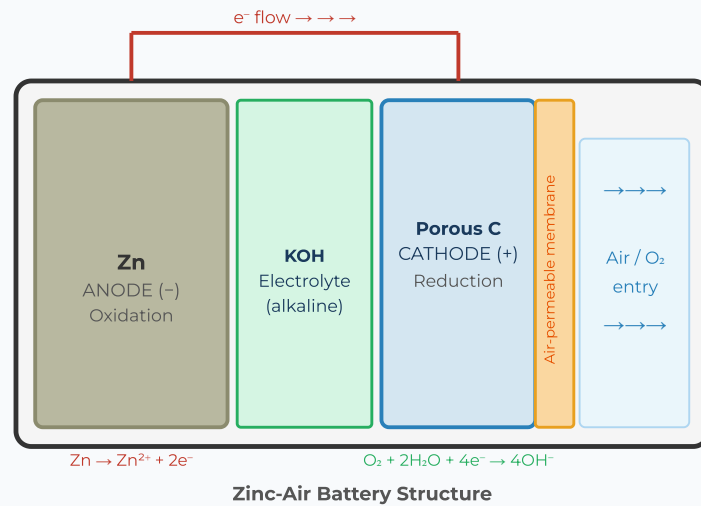


Fig. 14.1 – Zinc-Air Battery: Zn anode, alkaline KOH electrolyte, porous carbon air cathode

ZINC-AIR BATTERY REACTIONS

Anode (Oxidation): $\text{Zn} + 2\text{OH}^- \rightarrow \text{ZnO} + \text{H}_2\text{O} + 2\text{e}^-$ ($E^\circ = +1.25 \text{ V}$)

Cathode (Reduction): $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$ ($E^\circ = +0.40 \text{ V}$)

Overall: $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$ $E^\circ_{\text{cell}} \approx 1.65 \text{ V}$

Electrolyte: KOH (alkaline, 30-40%)

Energy density: 350-500 Wh/kg (very high!)

14.2 Lithium-Ion Battery (Secondary)

✓ WORKING PRINCIPLE

Li-ion batteries work by the **intercalation mechanism** — lithium ions move back and forth between the anode (graphite) and cathode (metal oxide) through a liquid electrolyte during charging and discharging. This is often called a "**rocking chair**" battery.

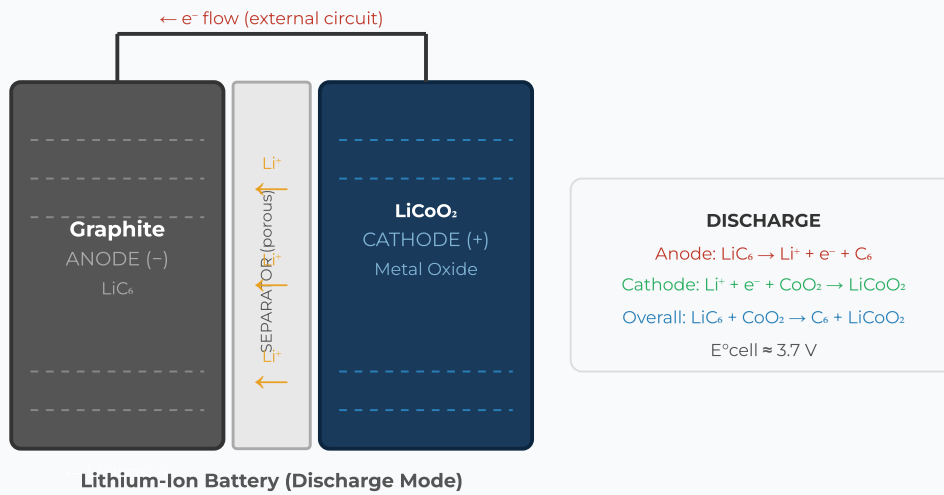


Fig. 14.2 – Lithium-Ion Battery: Li^+ migrate from anode (graphite) to cathode (LiCoO_2) during discharge

LITHIUM-ION BATTERY REACTIONS

Anode (Discharge): $\text{LiC}_6 \rightarrow 6\text{C} + \text{Li}^+ + \text{e}^-$

Cathode (Discharge): $\text{CoO}_2 + \text{Li}^+ + \text{e}^- \rightarrow \text{LiCoO}_2$

Charging: Reverse of above (using external power)

Cell voltage: $\sim 3.6\text{--}3.7\text{ V}$ (nominal)

Energy density: $150\text{--}250\text{ Wh/kg}$

Cycle life: $500\text{--}1500$ charge-discharge cycles

§15 Fuel Cells – H₂-O₂ & PEMFC

✓ STEP 1 – WHAT IS A FUEL CELL?

A **fuel cell** is an electrochemical device that continuously converts the chemical energy of a fuel (usually hydrogen) and an oxidant (oxygen/air) into electrical energy, *as long as fuel is supplied*. Unlike batteries, fuel cells do NOT store energy — they generate it continuously like power plants, but electrochemically.

Analogy: A fuel cell is like a car engine, but instead of burning fuel to create mechanical energy, it combines fuel and oxygen electrochemically to produce electricity. The only "exhaust" is water — completely clean!

Feature	Battery	Fuel Cell
Energy storage	Stores energy internally	Continuously uses external fuel
Refueling	Needs recharging	Refuel with H ₂ continuously
Byproduct	Heat (sometimes toxic)	Water only (H ₂ -O ₂ fuel cell)
Efficiency	60–90%	40–70% (can be higher with co-generation)
Applications	Portable electronics, EVs	Buses, ships, stationary power

15.1 Hydrogen-Oxygen Fuel Cell

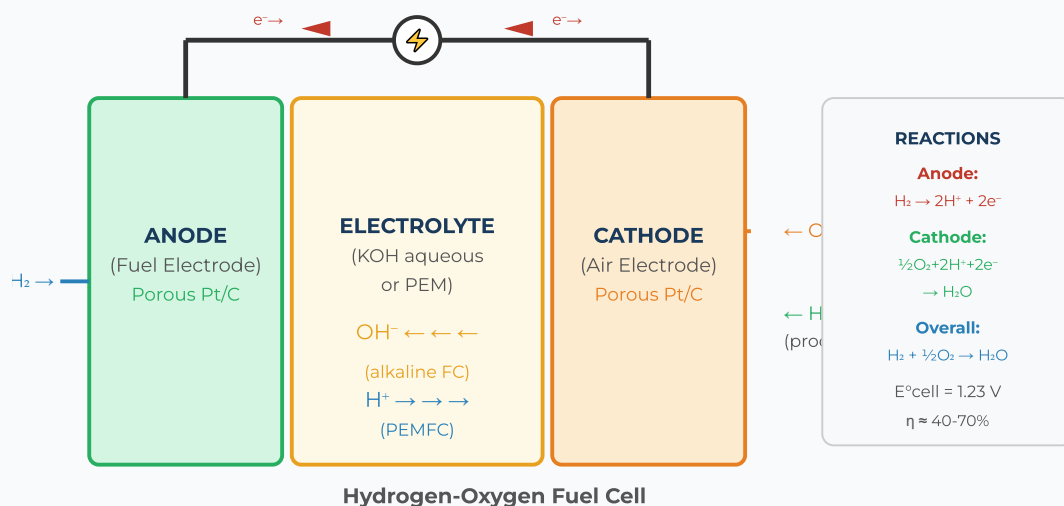
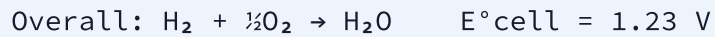


Fig. 15.1 – H₂-O₂ Fuel Cell: H₂ oxidized at anode, O₂ reduced at cathode, only H₂O formed

H₂-O₂ FUEL CELL REACTIONS



$$\Delta G^\circ = -nFE^\circ = -(2)(96500)(1.23) = -237,390 \text{ J/mol} \approx -237.4 \text{ kJ/mol}$$

15.2 Polymer Electrolyte Membrane (PEM) Fuel Cell

✓ WHAT MAKES PEMFC SPECIAL?

PEM fuel cells use a **solid polymer membrane** (like Nafion®) as the electrolyte instead of a liquid. This membrane conducts H⁺ ions (protons) but blocks electrons and gas crossover. They operate at low temperatures (60–90°C) and are ideal for automotive applications.

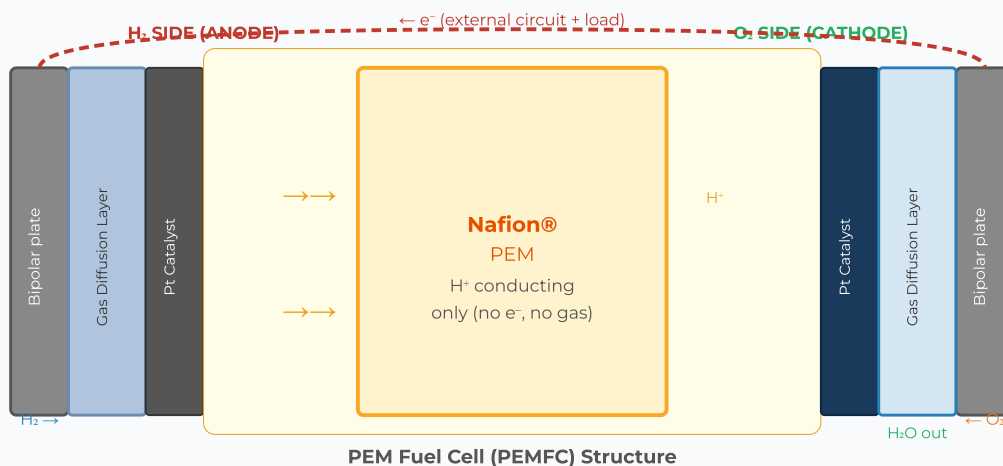


Fig. 15.2 – PEMFC: Nafion® membrane conducts H⁺ only; electrons flow through external circuit

15.3 Comparison – H₂-O₂ FC vs PEMFC

Feature	H ₂ -O ₂ (Alkaline FC)	PEMFC
Electrolyte	KOH solution (liquid)	Nafion® polymer membrane (solid)
Operating temp.	60–90°C	50–100°C
Ion transported	OH ⁻ (anions)	H ⁺ (protons)
Catalyst	Nickel or Pt	Pt/C nanoparticles
Start-up time	Slower	Fast (good for vehicles)
Applications	Space shuttle power	Fuel cell vehicles, portable power
Efficiency	60–70%	40–60%

§16 Engineering Applications of Electrochemistry

16.1 Applications in Various Engineering Fields

Field	Application	Principle Used
Electrical Engineering	Batteries for UPS, EVs, grid storage	Electrochemical energy storage
Civil Engineering	Corrosion protection (cathodic protection)	Galvanic cells, redox reactions
Chemical Engineering	Electrolysis, electroplating, electroforming	Electrolytic cells, Faraday's laws
Biomedical Engineering	Biosensors, pacemaker batteries	Amperometric sensors, electrochemistry
Environmental Engineering	Water treatment, heavy metal removal	Electrodialysis, electrocoagulation
Aerospace Engineering	Fuel cells for spacecraft, corrosion monitoring	H ₂ -O ₂ fuel cells
Materials Engineering	Electrodeposition, surface hardening	Electrolytic deposition
Petroleum Engineering	Corrosion in pipelines, cathodic protection	Galvanic protection

16.2 Corrosion and Its Prevention

✓ ELECTROCHEMISTRY OF CORROSION

Corrosion is essentially a **galvanic process**. When two dissimilar metals are in contact in the presence of an electrolyte, a galvanic cell forms. The more active metal (lower reduction potential) acts as anode and corrodes.

- Iron oxidation: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ (anode, corrodes)
- Oxygen reduction: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ (cathode)
- Rusting: $4\text{Fe}(\text{OH})_2 + \text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (rust)

16.3 Cathodic Protection

Method to prevent corrosion of iron/steel structures (bridges, ships, pipelines):

- **Sacrificial anode method:** Attach a more active metal (Zn, Mg) to iron. The active metal corrodes instead of iron.

- **Impressed current method:** Apply external current to make iron the cathode (prevents oxidation).

§17 Solved Problems (40+ Numerical & Conceptual)

17.1 EMF and Cell Potential Problems

PROBLEM 1 – BASIC EMF CALCULATION

Calculate the standard EMF of the cell: $\text{Zn} \mid \text{Zn}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$. Given: $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$, $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$

✓ **SOLUTION:**

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$E^\circ_{\text{cell}} = E^\circ(\text{Cu}^{2+}/\text{Cu}) - E^\circ(\text{Zn}^{2+}/\text{Zn})$$

$$E^\circ_{\text{cell}} = (+0.34) - (-0.76) = 0.34 + 0.76$$

✓ **$E^\circ_{\text{cell}} = +1.10 \text{ V}$ (Spontaneous — galvanic cell)**

PROBLEM 2 – IDENTIFY CATHODE AND ANODE

For the cell involving Fe^{2+}/Fe ($E^\circ = -0.44 \text{ V}$) and Cu^{2+}/Cu ($E^\circ = +0.34 \text{ V}$), identify cathode, anode, and find E°_{cell} .

✓ **SOLUTION:**

Higher E° = cathode → Cu^{2+}/Cu is the CATHODE (+0.34 V)

Lower E° = anode → Fe^{2+}/Fe is the ANODE (-0.44 V)

$$E^\circ_{\text{cell}} = 0.34 - (-0.44) = 0.34 + 0.44$$

✓ **$E^\circ_{\text{cell}} = +0.78 \text{ V}$, Cell: $\text{Fe} \mid \text{Fe}^{2+} \parallel \text{Cu}^{2+} \mid \text{Cu}$**

PROBLEM 3 – ΔG FROM EMF

Calculate ΔG° for the Daniell cell ($E^\circ_{\text{cell}} = 1.10 \text{ V}$, $n = 2$)

✓ **SOLUTION:**

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

$$\Delta G^\circ = -(2)(96500)(1.10) \text{ J/mol}$$

$$\Delta G^\circ = -212,300 \text{ J/mol}$$

✓ **$\Delta G^\circ = -212.3 \text{ kJ/mol}$ (negative $\rightarrow$ spontaneous ✓)**

PROBLEM 4 – EQUILIBRIUM CONSTANT FROM E°

Find the equilibrium constant K for the Daniell cell reaction at 298 K. ($E^\circ_{\text{cell}} = 1.10 \text{ V}$, $n = 2$)

✓ **SOLUTION:**

$$\log K = nE^\circ_{\text{cell}} / 0.0591$$

$$\log K = (2 \times 1.10) / 0.0591 = 2.20 / 0.0591$$

$$\log K = 37.23$$

✓ **$K = 10^{37.23} \approx 1.7 \times 10^{37}$ (reaction strongly favors products)**

PROBLEM 5 – SPONTANEITY CHECK

Is the reaction $\text{Ag} + \text{Cu}^{2+} \rightarrow \text{Ag}^+ + \text{Cu}$ spontaneous? [$E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$]

✓ **SOLUTION:**

$$\text{If Ag is anode: } E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.34 - 0.80 = -0.46 \text{ V}$$

$$E^\circ_{\text{cell}} = -0.46 \text{ V (negative!)}$$

✓ **Non-spontaneous in this direction. Reverse is spontaneous: $\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$**

17.2 Nernst Equation Problems

PROBLEM 6 – NERNST EQUATION CALCULATION

Calculate the EMF of the Daniell cell at 298 K when $[\text{Zn}^{2+}] = 0.1 \text{ M}$ and $[\text{Cu}^{2+}] = 0.01 \text{ M}$. ($E^\circ_{\text{cell}} = 1.10 \text{ V}$)

✓ **SOLUTION:**

Cell reaction: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$ ($n = 2$)

$Q = [\text{Zn}^{2+}] / [\text{Cu}^{2+}] = 0.1 / 0.01 = 10$

$E = E^\circ - (0.0591/n) \times \log Q$

$E = 1.10 - (0.0591/2) \times \log(10)$

$E = 1.10 - (0.02955)(1) = 1.10 - 0.02955$

✓ **$E = 1.070 \text{ V}$**

PROBLEM 7 – CONCENTRATION CELL EMF

Calculate the EMF of a concentration cell: $\text{Cu} \mid \text{Cu}^{2+}(0.001 \text{ M}) \parallel \text{Cu}^{2+}(0.1 \text{ M}) \mid \text{Cu}$ at 298 K

✓ **SOLUTION:**

Both electrodes: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$, so $E^\circ = 0$

$Q = [\text{Cu}^{2+} \text{ at anode}] / [\text{Cu}^{2+} \text{ at cathode}] = 0.001 / 0.1 = 0.01$

$E = 0 - (0.0591/2) \times \log(0.01)$

$E = -(0.02955) \times (-2) = +0.059 \text{ V}$

✓ **$E = +0.059 \text{ V}$ (Dilute side = anode, Concentrated side = cathode)**

PROBLEM 8 – NERNST FOR HYDROGEN ELECTRODE

Calculate the potential of a hydrogen electrode at pH = 3, $P(\text{H}_2) = 1 \text{ atm}$, $T = 298 \text{ K}$

✓ **SOLUTION:**

Electrode: $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ ($n = 2$)

$$E = E^\circ - (0.0591/2) \times \log(P(\text{H}_2)/[\text{H}^+]^2)$$

At pH = 3: $[\text{H}^+] = 10^{-3} \text{ M}$

$$E = 0 - (0.02955) \times \log(1/(10^{-3})^2) = -0.02955 \times \log(10^6)$$

$$E = -0.02955 \times 6 = -0.1773 \text{ V}$$

✓ **$E = -0.177 \text{ V}$ [or simply: $E = -0.0591 \times \text{pH} = -0.0591 \times 3 = -0.177 \text{ V}$]**

PROBLEM 9 – EMF AND TEMPERATURE

The EMF of a cell is 0.90 V at 298 K. If $Q = 100$ and $n = 2$, find E°_{cell} .

✓ **SOLUTION:**

$$E = E^\circ - (0.0591/n) \times \log Q$$

$$0.90 = E^\circ - (0.0591/2) \times \log(100)$$

$$0.90 = E^\circ - (0.02955)(2)$$

$$0.90 = E^\circ - 0.059$$

$$E^\circ = 0.90 + 0.059 = 0.959 \text{ V}$$

✓ **$E^\circ_{\text{cell}} = 0.959 \text{ V} \approx 0.96 \text{ V}$**

PROBLEM 10 – FIND $[Zn^{2+}]$ FROM EMF

The EMF of a Daniel cell is 1.07 V at 298 K with $[Cu^{2+}] = 1.0$ M. Find $[Zn^{2+}]$. ($E^\circ_{\text{cell}} = 1.10$ V)

✓ SOLUTION:

$$E = E^\circ - (0.0591/2) \times \log([Zn^{2+}]/[Cu^{2+}])$$

$$1.07 = 1.10 - (0.02955) \times \log([Zn^{2+}]/1.0)$$

$$-0.03 = -0.02955 \times \log[Zn^{2+}]$$

$$\log[Zn^{2+}] = 0.03/0.02955 = 1.015$$

$$[Zn^{2+}] = 10^{1.015} \approx 10.35 \text{ M}$$

✓ $[Zn^{2+}] \approx 10.35$ M (high Zn^{2+} reduces EMF as expected)

17.3 Conductivity Problems

PROBLEM 11 – SPECIFIC CONDUCTANCE

A conductivity cell with cell constant 0.80 cm^{-1} shows a resistance of 400Ω . Find the specific conductance.

✓ **SOLUTION:**

$$\text{Conductance } G = 1/R = 1/400 = 0.0025 \text{ S}$$

$$\kappa = G \times \text{Cell constant} = 0.0025 \times 0.80$$

$$\checkmark \kappa = 0.002 \text{ S/cm} = 2 \times 10^{-3} \text{ S cm}^{-1}$$

PROBLEM 12 – MOLAR CONDUCTANCE

The specific conductance of 0.01 M KCl solution is $1.47 \times 10^{-3} \text{ S/cm}$. Find molar conductance Λ_m .

✓ **SOLUTION:**

$$\Lambda_m = (\kappa \times 1000) / C$$

$$\Lambda_m = (1.47 \times 10^{-3} \times 1000) / 0.01$$

$$\Lambda_m = 1.47 / 0.01 = 147 \text{ S cm}^2/\text{mol}$$

$$\checkmark \Lambda_m = 147 \text{ S cm}^2 \text{ mol}^{-1}$$

PROBLEM 13 – CELL CONSTANT FROM KNOWN SOLUTION

A conductivity cell with 0.01 M KCl ($\kappa = 1.29 \times 10^{-3} \text{ S/cm}$) shows resistance of 400Ω . Find cell constant.

✓ **SOLUTION:**

$$G = 1/R = 1/400 = 2.5 \times 10^{-3} \text{ S}$$

$$K_{\text{cell}} = \kappa / G = (1.29 \times 10^{-3}) / (2.5 \times 10^{-3})$$

$$\checkmark K_{\text{cell}} = 0.516 \text{ cm}^{-1}$$

PROBLEM 14 – DEGREE OF DISSOCIATION

Λ_m of 0.1 M $\text{CH}_3\text{COOH} = 5.2 \text{ S cm}^2/\text{mol}$. Λ°_m of $\text{CH}_3\text{COOH} = 390.7 \text{ S cm}^2/\text{mol}$. Find degree of dissociation α .

✓ SOLUTION:

$$\alpha = \Lambda_m / \Lambda^\circ_m$$

$$\alpha = 5.2 / 390.7$$

$$\checkmark \alpha = 0.0133 = 1.33\% \text{ (weak acid — very low dissociation)}$$

PROBLEM 15 – KOHLRAUSCH LAW CALCULATION

Using Kohlrausch's law, calculate Λ°_m of CH_3COOH . Given: $\Lambda^\circ_m(\text{CH}_3\text{COONa}) = 91 \text{ S cm}^2/\text{mol}$, $\Lambda^\circ_m(\text{HCl}) = 426 \text{ S cm}^2/\text{mol}$, $\Lambda^\circ_m(\text{NaCl}) = 126 \text{ S cm}^2/\text{mol}$

✓ SOLUTION:

$$\Lambda^\circ_m(\text{CH}_3\text{COOH}) = \Lambda^\circ_m(\text{CH}_3\text{COONa}) + \Lambda^\circ_m(\text{HCl}) - \Lambda^\circ_m(\text{NaCl})$$

$$= 91 + 426 - 126$$

$$\checkmark \Lambda^\circ_m(\text{CH}_3\text{COOH}) = 391 \text{ S cm}^2 \text{ mol}^{-1}$$

PROBLEM 16 – DISSOCIATION CONSTANT K_a FROM CONDUCTANCE

For 0.1 M CH_3COOH : $\alpha = 0.0133$ (from Problem 14). Find K_a .

✓ SOLUTION:

$$K_a = C\alpha^2 / (1 - \alpha) \approx C\alpha^2 \text{ (since } \alpha \ll 1)$$

$$K_a = 0.1 \times (0.0133)^2 = 0.1 \times 1.769 \times 10^{-4}$$

$$\checkmark K_a = 1.769 \times 10^{-5} \text{ mol/L } (\approx \text{ literature value } 1.8 \times 10^{-5} \checkmark)$$

17.4 Faraday's Laws Problems

PROBLEM 17 – FIRST LAW OF FARADAY

How many grams of copper are deposited when 2 amperes of current flow for 2 hours through CuSO_4 solution? (At. wt Cu = 63.5, $n = 2$)

✓ **SOLUTION:**

$$Q = I \times t = 2 \times (2 \times 3600) = 2 \times 7200 = 14400 \text{ C}$$

$$\text{Equivalent weight of Cu} = 63.5/2 = 31.75 \text{ g/eq}$$

$$\text{mass} = (Q \times \text{Eq. wt}) / F = (14400 \times 31.75) / 96500$$

$$\text{mass} = 457,200 / 96500 = 4.737 \text{ g}$$

✓ **Mass of Cu deposited = 4.74 g**

PROBLEM 18 – TIME REQUIRED FOR DEPOSITION

How long will it take to deposit 10 g of silver from AgNO_3 solution using 3 A current? (At. wt Ag = 108, $n = 1$)

✓ **SOLUTION:**

$$\text{moles of Ag} = 10/108 = 0.09259 \text{ mol}$$

$$\text{Charge needed} = n \times F \times \text{moles} = 1 \times 96500 \times 0.09259 = 8935 \text{ C}$$

$$\text{Time} = Q/I = 8935/3 = 2978 \text{ s}$$

✓ **Time = 2978 s $\approx$ 49.6 minutes**

PROBLEM 19 – CURRENT EFFICIENCY

In an electrolysis experiment, 3.00 g of Cu is deposited when 5 A flows for 1800 s. Find current efficiency. (At. wt Cu = 63.5, $n = 2$)

✓ **SOLUTION:**

$$\text{Theoretical mass} = (I \times t \times M) / (n \times F) = (5 \times 1800 \times 63.5) / (2 \times 96500)$$

$$= 571500 / 193000 = 2.961 \text{ g}$$

$$\text{Current efficiency} = (\text{Actual/Theoretical}) \times 100 = (3.00/2.961) \times 100$$

✓ **Current efficiency = 101.3% $\approx$ 100% (slight experimental error)**

PROBLEM 20 – SECOND LAW OF FARADAY

The same quantity of charge deposits 1.08 g of Ag from AgNO_3 . How much Cu deposits from CuSO_4 ? (Eq. wt Ag = 108, Eq. wt Cu = 31.75)

✓ SOLUTION:

Faraday's 2nd Law: $m_1/m_2 = \text{Eq.wt}_1/\text{Eq.wt}_2$

$$1.08/m(\text{Cu}) = 108/31.75$$

$$m(\text{Cu}) = 1.08 \times 31.75/108 = 0.3175 \text{ g}$$

✓ Mass of Cu deposited = 0.317 g

17.5 Battery and Fuel Cell Problems**PROBLEM 21 – ZINC-AIR BATTERY CAPACITY**

A zinc-air button cell contains 0.5 g of Zn. Calculate the theoretical charge (capacity) in Coulombs and mAh. (At. wt Zn = 65.4, $n = 2$)

✓ SOLUTION:

$$\text{moles of Zn} = 0.5/65.4 = 7.645 \times 10^{-3} \text{ mol}$$

$$\text{Charge } Q = n \times F \times \text{moles} = 2 \times 96500 \times 7.645 \times 10^{-3} = 1475.5 \text{ C}$$

$$\text{Capacity in mAh} = Q(\text{C}) / 3.6 = 1475.5 / 3.6 = 409.9 \text{ mAh}$$

✓ Charge = 1475.5 C $\approx$ 410 mAh

PROBLEM 22 – ΔG FOR $\text{H}_2\text{-O}_2$ FUEL CELL

Calculate ΔG° for $\text{H}_2\text{-O}_2$ fuel cell at 298 K. ($E^\circ_{\text{cell}} = 1.23 \text{ V}$, $n = 2$)

✓ SOLUTION:

$$\Delta G^\circ = -nFE^\circ = -(2)(96500)(1.23)$$

$$= -237,390 \text{ J/mol}$$

✓ $\Delta G^\circ = -237.4 \text{ kJ/mol}$ (Gibbs energy of formation of H_2O ✓)

PROBLEM 23 – FUEL CELL EFFICIENCY

A hydrogen fuel cell produces 0.70 V at working condition. $E^{\circ}_{\text{theoretical}} = 1.23$ V. Calculate thermodynamic efficiency.

✓ **SOLUTION:**

$$\text{Efficiency} = (\text{Actual voltage} / \text{Theoretical voltage}) \times 100$$

$$\text{Efficiency} = (0.70 / 1.23) \times 100 = 56.9\%$$

✓ **Efficiency $\approx$ 57% (much better than combustion engines $\sim$ 25%)**

PROBLEM 24 – LI-ION BATTERY ENERGY

A Li-ion cell has OCV = 3.7 V and capacity = 2600 mAh. Find total energy stored in Wh and Joules.

✓ **SOLUTION:**

$$\text{Energy (Wh)} = \text{Voltage} \times \text{Capacity} = 3.7 \times 2.6 \text{ Ah} = 9.62 \text{ Wh}$$

$$\text{Energy (J)} = \text{Wh} \times 3600 = 9.62 \times 3600 = 34,632 \text{ J}$$

✓ **Energy = 9.62 Wh = 34.6 kJ**

17.6 Sensor and Potentiometry Problems

PROBLEM 25 – PH FROM ELECTRODE POTENTIAL

A glass electrode has $E_{\text{glass}} = 0.380$ V and $E_{\text{SCE}} = +0.242$ V. $E^{\circ}_{\text{glass}} = 0.50$ V. Find pH.

✓ **SOLUTION:**

$$E_{\text{cell}} = E_{\text{glass}} - E_{\text{SCE}} = 0.380 - 0.242 = 0.138 \text{ V}$$

$$E_{\text{glass}} = E^{\circ}_{\text{glass}} - 0.0591 \times \text{pH}$$

$$0.380 = 0.50 - 0.0591 \times \text{pH}$$

$$0.0591 \times \text{pH} = 0.50 - 0.380 = 0.120$$

$$\text{pH} = 0.120 / 0.0591$$

✓ **pH = 2.03 (acidic solution)**

PROBLEM 26 – ISE SENSITIVITY

A Ca^{2+} ISE ($z = 2$) shows a potential change of 30 mV when Ca^{2+} concentration changes from 0.001 M to 0.01 M. What is the theoretical Nernstian response?

✓ SOLUTION:

$$\text{Theoretical } \Delta E = (0.0591/z) \times \log([C_2]/[C_1])$$

$$= (0.0591/2) \times \log(0.01/0.001) = (0.02955) \times \log(10)$$

$$= 0.02955 \times 1 = 29.55 \text{ mV} \approx 29.6 \text{ mV}$$

✓ Theoretical $\Delta E = 29.6 \text{ mV/decade}$ (measured $\approx 30 \text{ mV}$ — near-ideal Nernstian!)

PROBLEM 27 – POTENTIOMETRIC TITRATION EP DETECTION

In a Fe^{2+} vs KMnO_4 titration, the following data is observed. Find the equivalence point volume: $V(\text{mL})$: 10, 12, 13, 13.5, 14, 15, 16 | $E(\text{mV})$: 200, 230, 280, 350, 750, 900, 950

✓ SOLUTION:

Calculate $\Delta E/\Delta V$ for each interval:

$$10 \rightarrow 12: \Delta E/\Delta V = 30/2 = 15 \text{ mV/mL}$$

$$12 \rightarrow 13: \Delta E/\Delta V = 50/1 = 50 \text{ mV/mL}$$

$$13 \rightarrow 13.5: \Delta E/\Delta V = 70/0.5 = 140 \text{ mV/mL}$$

$$13.5 \rightarrow 14: \Delta E/\Delta V = 400/0.5 = 800 \text{ mV/mL} \leftarrow \text{Maximum!}$$

$$14 \rightarrow 15: \Delta E/\Delta V = 150/1 = 150 \text{ mV/mL}$$

✓ EP $\approx 13.75 \text{ mL}$ (midpoint of 13.5 $\rightarrow$ 14 where $\Delta E/\Delta V$ is maximum)

PROBLEM 28 – CONDUCTOMETRIC TITRATION EQUIVALENCE POINT

During the titration of HCl with NaOH, conductance values were: $V = 0, 5, 10, 15, 20, 25$ mL; $G = 800, 600, 400, 200, 400, 600$ μS . Find equivalence point volume.

✓ SOLUTION:

Minimum conductance occurs at $V = 15$ mL ($G = 200$ μS)

Before 15 mL: G decreases (H^+ replaced by Na^+)

After 15 mL: G increases (excess NaOH adds OH^-)

✓ Equivalence point volume = 15 mL (at minimum conductance)

PROBLEM 29 – GLUCOSE SENSOR AMPEROMETRIC RESPONSE

A glucose biosensor produces 2 μA current for 1 mM glucose and 8 μA for 4 mM glucose. Is the response linear? What current is expected for 3 mM?

✓ SOLUTION:

Check linearity: $i/C = 2/1 = 2$ $\mu\text{A}/\text{mM}$ for 1 mM

$i/C = 8/4 = 2$ $\mu\text{A}/\text{mM}$ for 4 mM

Sensitivity = 2 $\mu\text{A}/\text{mM}$ (constant $\rightarrow$ linear response ✓)

At 3 mM: $i = 2 \times 3 = 6$ μA

✓ Linear amperometric response. At 3 mM $\rightarrow i = 6$ μA

PROBLEM 30 – NERNST EQUATION FOR Cl^- ELECTRODE

Find the potential of Cl^- ISE when $[\text{Cl}^-] = 0.05$ M. $E^\circ = +0.22$ V vs SHE, $z = -1$.

✓ SOLUTION:

For anion ISE: $E = E^\circ + (0.0591/z) \times \log[\text{Cl}^-] \rightarrow z = -1$ for Cl^-

$E = 0.22 + (0.0591/-1) \times \log(0.05)$

$E = 0.22 - 0.0591 \times (-1.301)$

$E = 0.22 + 0.0769$

✓ $E = +0.297$ V

PROBLEM 31 – GALVANIC CELL VS ELECTROLYTIC

A cell has $E^\circ_{\text{cell}} = -0.50 \text{ V}$. Can it work as a galvanic cell? What voltage must be applied to make it electrolytic?

✓ **SOLUTION:**

$E^\circ_{\text{cell}} = -0.50 \text{ V}$ (negative) → Reaction is NON-spontaneous

→ Cannot work as a galvanic cell

For electrolysis: minimum voltage needed = $|E^\circ_{\text{cell}}| = 0.50 \text{ V}$

Actual applied voltage must be slightly higher than 0.50 V to overcome overpotential

✓ **Not galvanic. Need at least 0.50 V applied for electrolysis.**

PROBLEM 32 – SPECIFIC CONDUCTANCE FROM TWO READINGS

If $R_1 = 100 \, \Omega$ for 0.01 M KCl ($\kappa = 1.41 \times 10^{-3} \text{ S/cm}$) and $R_2 = 250 \, \Omega$ for unknown solution, find κ of unknown.

✓ **SOLUTION:**

Cell constant $K_c = \kappa_1 \times R_1 = 1.41 \times 10^{-3} \times 100 = 0.141 \text{ cm}^{-1}$

$\kappa_2 = K_c / R_2 = 0.141 / 250$

✓ **κ (unknown) = $5.64 \times 10^{-4} \text{ S cm}^{-1}$**

PROBLEM 33 – OVERVOLTAGE / OVERPOTENTIAL

Theoretically, water electrolysis requires 1.23 V , but in practice 1.70 V is needed. Calculate the overpotential.

✓ **SOLUTION:**

Overpotential = Practical voltage - Theoretical voltage

$= 1.70 - 1.23 = 0.47 \text{ V}$

✓ **Overpotential = 0.47 V (due to activation energy barriers at electrodes)**

PROBLEM 34 – ELECTROPLATING MASS

How long must a 2 A current flow to deposit 1.0 g of Cr from Cr^{3+} solution? (At. wt Cr = 52, $n = 3$)

✓ **SOLUTION:**

$$\text{Eq. wt. of Cr} = 52/3 = 17.33 \text{ g/eq}$$

$$Q = (\text{mass} \times F) / \text{Eq.wt} = (1.0 \times 96500) / 17.33 = 5569.2 \text{ C}$$

$$t = Q / I = 5569.2 / 2 = 2784.6 \text{ s}$$

✓ **Time = 2784.6 s $\approx$ 46.4 minutes**

PROBLEM 35 – SALT BRIDGE SELECTION

[Conceptual] Why is KNO_3 preferred over KCl as a salt bridge when one half-cell contains Ag^+ ions?

✓ **SOLUTION:**

If KCl is used, Cl^- from salt bridge migrates into Ag^+ half-cell



This changes Ag^+ concentration and creates a junction potential

KNO_3 is used instead: NO_3^- does NOT precipitate with Ag^+

✓ **KNO_3 is used to avoid AgCl precipitation which would disturb the cell.**

PROBLEM 36 – MEMBRANE POTENTIAL IN ISE

A pH glass electrode is calibrated: in buffer of pH 4 it shows $E = 0.190 \text{ V}$; in buffer of pH 7, $E = 0.013 \text{ V}$. Find the slope (mV/pH unit).

✓ **SOLUTION:**

$$\Delta E = 0.190 - 0.013 = 0.177 \text{ V} = 177 \text{ mV}$$

$$\Delta \text{pH} = 7 - 4 = 3 \text{ pH units}$$

$$\text{Slope} = \Delta E / \Delta \text{pH} = 177 / 3 = 59 \text{ mV/pH unit}$$

✓ **Slope = 59 mV/pH unit (ideal Nernstian = 59.1 mV/pH — excellent!)**

PROBLEM 37 – LITHIUM-ION ENERGY DENSITY

A Li-ion battery pack: 100 cells in series, each 3.7 V, 2000 mAh. Find total voltage, total energy (Wh), and energy density if pack weighs 2 kg.

✓ SOLUTION:

$$\text{Total voltage} = 100 \times 3.7 = 370 \text{ V}$$

$$\text{Energy} = V \times \text{Ah} = 370 \times 2.0 = 740 \text{ Wh}$$

$$\text{Energy density} = 740 \text{ Wh} / 2 \text{ kg} = 370 \text{ Wh/kg}$$

✓ Total E = 740 Wh; Energy density = 370 Wh/kg

PROBLEM 38 – MOLAR CONDUCTANCE OF WEAK ELECTROLYTE

$\Lambda^\circ \text{m}$ of $\text{NH}_4\text{OH} = 238.3 \text{ S cm}^2/\text{mol}$. Λm at 0.01 M = $9.5 \text{ S cm}^2/\text{mol}$. Find degree of dissociation and K_a .

✓ SOLUTION:

$$\alpha = \Lambda \text{m} / \Lambda^\circ \text{m} = 9.5 / 238.3 = 0.0399 \approx 0.04$$

$$K_a = C\alpha^2 / (1-\alpha) = (0.01 \times 0.04^2) / (1 - 0.04)$$

$$= (0.01 \times 0.0016) / 0.96 = 1.6 \times 10^{-5} / 0.96$$

✓ $\alpha = 4.0\%$, $K_a = 1.67 \times 10^{-5}$ (close to K_b of $\text{NH}_3 = 1.8 \times 10^{-5}$ ✓)

PROBLEM 39 – NUMBER OF ELECTRONS FROM EMF AND ΔG

ΔG° for a cell = -251.9 kJ/mol and $E^\circ = 1.30 \text{ V}$. Find the number of electrons transferred.

✓ SOLUTION:

$$\Delta G^\circ = -nFE^\circ$$

$$-251,900 \text{ J/mol} = -n \times 96500 \times 1.30$$

$$n = 251900 / (96500 \times 1.30) = 251900 / 125450$$

$$n = 2.007 \approx 2$$

✓ $n = 2$ electrons transferred

PROBLEM 40 – PEMFC POWER OUTPUT

A PEMFC single cell produces 0.70 V at 1.5 A. Find: (a) Power output (b) Moles of H₂ consumed per second (n=2)

✓ SOLUTION:

$$(a) \text{ Power} = V \times I = 0.70 \times 1.5 = 1.05 \text{ W}$$

$$(b) \text{ Current} = 1.5 \text{ A} = 1.5 \text{ C/s}$$

$$\text{Moles of H}_2/\text{s} = I / (n \times F) = 1.5 / (2 \times 96500) = 7.77 \times 10^{-6} \text{ mol/s}$$

✓ Power = 1.05 W; H₂ consumed = 7.77×10^{-6} mol/s = 7.77 μmol/s

PROBLEM 41 – NERNST AT NON-STANDARD TEMPERATURE

Calculate E_{cell} for Zn|Zn²⁺(0.001M)||Cu²⁺(0.1M)|Cu at 60°C (333 K). E°_{cell} = 1.10 V, n=2

✓ SOLUTION:

$$Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.001/0.1 = 0.01$$

$$RT/F \text{ at } 333\text{K} = (8.314 \times 333) / 96500 = 0.02868 \text{ V}$$

$$E = E^\circ - (2.303RT/nF) \times \log Q = 1.10 - (2.303 \times 0.02868/2) \times \log(0.01)$$

$$= 1.10 - (0.03302) \times (-2) = 1.10 + 0.0660$$

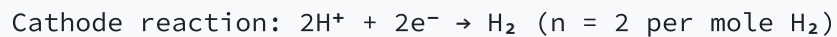
✓ E_{cell} = 1.166 V at 60°C (higher than at 25°C because Q < 1)

PROBLEM 42 – H₂ PRODUCTION IN ELECTROLYSIS

Calculate volume of H₂ gas (STP) produced when 2 A current flows for 1 hour through water. (Molar vol = 22.4 L/mol)

✓ SOLUTION:

$$Q = 2 \times 3600 = 7200 \text{ C}$$



$$\text{Moles of H}_2 = Q/(n \times F) = 7200/(2 \times 96500) = 0.03731 \text{ mol}$$

$$\text{Volume at STP} = 0.03731 \times 22.4 = 0.836 \text{ L}$$

✓ Volume of H₂ = 0.836 L = 836 mL at STP

§18 Questions & Answers (2M & 5M)

18.1 Two-Mark Questions and Answers

2M Q1. Define EMF of an electrochemical cell.

The EMF (Electromotive Force) of an electrochemical cell is the maximum potential difference between the two electrodes when no current is flowing through the external circuit. It is measured in volts (V) and given by: $E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$. A positive EMF indicates a spontaneous reaction.

2M Q2. What is the Nernst equation and when is it used?

The Nernst equation gives the cell potential at non-standard conditions: $E = E^{\circ} - (0.0591/n) \times \log Q$ at 298 K, where Q is the reaction quotient. It is used when ion concentrations differ from 1 M, or at non-standard temperature and pressure, to calculate actual cell potential.

2M Q3. State Kohlrausch's Law of Independent Migration of Ions.

Kohlrausch's Law states that at infinite dilution, each ion migrates independently of the other ions and contributes a fixed, characteristic value to the total molar conductance of an electrolyte: $\Lambda^{\circ}m = \lambda^{\circ}(\text{cation}) + \lambda^{\circ}(\text{anion})$. This law is used to find limiting molar conductance of weak electrolytes.

2M Q4. What is the function of a salt bridge?

A salt bridge (U-tube filled with KCl or KNO_3 in agar gel) serves two functions: (1) It maintains electrical neutrality in both half-cells by allowing migration of ions (K^+ migrates to cathode, $\text{Cl}^-/\text{NO}_3^-$ migrates to anode), and (2) It completes the electrical circuit without mixing the two electrode solutions.

2M Q5. Define specific conductance and molar conductance.

Specific conductance (κ): Conductance of a cube of electrolyte solution of unit dimensions ($1\text{ cm} \times 1\text{ cm} \times 1\text{ cm}$); $\kappa = G \times (l/A)$ [S/cm]. **Molar conductance (Λ_m):** Conductance of a solution containing 1 mole of electrolyte; $\Lambda_m = (\kappa \times 1000)/C$ [$\text{S cm}^2/\text{mol}$], where C is concentration in mol/L.

2M Q6. Distinguish between galvanic and electrolytic cells.

A **galvanic cell** converts chemical energy to electrical energy via a spontaneous redox reaction ($\Delta G < 0$, $E > 0$). The anode is negative. Examples: batteries. An **electrolytic cell** uses electrical energy to drive a non-spontaneous reaction ($\Delta G > 0$). The anode is positive. Examples: electroplating, electrolysis of water.

2M Q7. What is a potentiometric sensor? Give one example.

A potentiometric sensor measures the open-circuit potential (voltage) between an ion-selective electrode and a reference electrode. The potential varies with ion activity according to the Nernst equation: $E = \text{const} + (0.0591/z) \times \log[\text{ion}]$. Example: **Glass electrode (pH meter)** — measures H^+ activity, giving 59.1 mV per pH unit change.

2M Q8. Define conductometric titration and name the equivalence point feature.

Conductometric titration is an analytical technique where the conductance of the titrand solution is measured as titrant is added. Since different ions have different conductances, the measured conductance changes with addition. The **equivalence point** is indicated by a **minimum conductance** (for strong acid–strong base) or by a distinct kink in the conductance vs. volume curve.

2M Q9. What is a fuel cell? How is it different from a battery?

A **fuel cell** is an electrochemical device that continuously converts chemical energy of an externally supplied fuel (H_2) and oxidant (O_2) to electricity. Unlike a **battery** (which stores limited chemical energy internally), a fuel cell operates continuously as long as fuel is supplied. It does not need recharging — only refueling.

2M Q10. Write the reactions at each electrode in a H_2 - O_2 fuel cell.

Anode (oxidation): $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$ (or $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$ in alkaline)

Cathode (reduction): $\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$

Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$, $E^\circ_{\text{cell}} = 1.23 \text{ V}$. The only product is water — making it zero-emission.

18.2 Five-Mark Questions and Answers

5M

Q11. Derive the Nernst equation and explain its significance with an example.

Derivation:

Starting from thermodynamics: $\Delta G = \Delta G^\circ + RT \ln Q$... (1)

Since $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$, substituting in (1):

$$-nFE = -nFE^\circ + RT \ln Q$$

Dividing by $-nF$: $E = E^\circ - (RT/nF) \ln Q$

At 298 K, converting to $\log_{10}$: $E = E^\circ - (0.0591/n) \log Q$

Terms: E° = standard potential (V), n = electrons transferred, Q = [products]/[reactants]

Significance: (1) Calculates actual cell EMF at any concentration. (2) Explains pH measurement via glass electrode. (3) Shows that increasing product concentration decreases EMF. (4) At equilibrium: $E = 0$ and $Q = K$, giving $E^\circ = (0.0591/n) \log K$.

Example: For Daniell cell with $[Zn^{2+}] = 0.1 \text{ M}$, $[Cu^{2+}] = 1 \text{ M}$:

$$Q = 0.1/1 = 0.1; E = 1.10 - (0.0591/2) \times \log(0.1) = 1.10 + 0.0296 = 1.13 \text{ V (higher than } E^\circ\text{)}$$

5M

Q12. Explain the working of a Lithium-ion battery with diagram, reactions, and applications.

Structure: Graphite (anode) | $LiPF_6$ in organic solvent (electrolyte) | $LiCoO_2$ (cathode)

Discharge Reactions:

Anode: $LiC_6 \rightarrow Li^+ + e^- + C_6$ (Li^+ deintercalates from graphite)

Cathode: $CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$ (Li^+ intercalates into CoO_2)

Overall: $LiC_6 + CoO_2 \rightarrow C_6 + LiCoO_2$; $E \approx 3.7 \text{ V}$

Charging: Reverse reactions driven by external power supply.

Key features: High voltage (3.6–3.7 V), high energy density (150–250 Wh/kg), no memory effect, 500–1500 cycles, lightweight.

Applications: Smartphones, laptops, electric vehicles, drones, grid energy storage, medical implants.

Principle: The cell potential (E) is monitored as titrant is added. The equivalence point is detected by a sudden sharp change in E.

Setup: Indicator electrode + Reference electrode (SCE) dipped in analyte solution, connected to a high-impedance voltmeter. Titrant added from burette.

Procedure: Add titrant in small increments → record E after each addition → plot E vs Volume.

EP Detection methods:

- (1) Inflection point on E vs V curve (steepest part)
- (2) Maximum of $\Delta E/\Delta V$ vs V (first derivative method)
- (3) Zero crossing of $\Delta^2 E/\Delta V^2$ (second derivative method — most precise)

Advantages: Applicable to colored solutions, no indicator needed, very accurate, can detect multiple EP.

Application: Fe^{2+} vs KMnO_4 titration (Pt indicator electrode, SCE reference).

Construction:

- **Anode:** Porous carbon paper + Pt catalyst (H_2 oxidation)
- **Cathode:** Porous carbon paper + Pt catalyst (O_2 reduction)
- **Membrane:** Nafion[®] — solid polymer, conducts H^+ only (not electrons or gases)
- **Gas diffusion layers (GDL)** on both sides for uniform gas distribution
- **Bipolar plates** for current collection and gas distribution

Working:

$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$ (anode, Pt catalyst)

H^+ ions pass through Nafion[®] membrane to cathode.

Electrons travel through external circuit (producing electricity).

At cathode: $\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O}$

Advantages: Low temperature operation (60–90°C), solid electrolyte (safe), fast startup, high power density, zero emissions (only H_2O formed).

Applications: Toyota Mirai, Hyundai Nexo (fuel cell vehicles), portable power, submarine power.

Principle: Conductance changes as ions are added or removed.

Initial solution: HCl gives H^+ and Cl^- . H^+ has very high mobility ($\lambda = 350 \text{ S cm}^2/\text{mol}$).

Region 1 – Before Equivalence Point: As NaOH is added, $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$. Fast H^+ ions replaced by slow Na^+ ions ($\lambda = 50 \text{ S cm}^2/\text{mol}$). Conductance DECREASES linearly.

At Equivalence Point: All H^+ consumed. Solution = NaCl only. MINIMUM conductance.

Region 2 – After Equivalence Point: Excess NaOH adds Na^+ and OH^- ($\lambda(\text{OH}^-) = 198 \text{ S cm}^2/\text{mol}$). Conductance INCREASES sharply.

Curve shape: V-shaped with minimum at EP. The EP volume = intersection point of the two straight-line regions.

Advantages over visual indicator: Works with colored solutions, highly precise, can detect multiple EPs.

§19 Quick Revision Sheet

⚡ Core EMF Formulas

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$$

$$\Delta G^{\circ} = -nFE^{\circ}$$

$$\log K_{\text{eq}} = nE^{\circ}/0.0591$$

$$\text{Nernst: } E = E^{\circ} - (0.0591/n) \log Q$$

$$\text{At equilibrium: } E = 0, Q = K$$

$$F = 96500 \text{ C/mol}$$

🧪 Conductivity Formulas

$$G = 1/R \text{ [Siemens]}$$

$$\kappa = G \times K_{\text{cell}} \text{ [S/cm]}$$

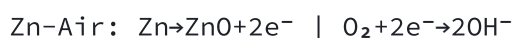
$$K_{\text{cell}} = l/A \text{ [cm}^{-1}\text{]}$$

$$\Lambda_m = (\kappa \times 1000)/C \text{ [S cm}^2\text{/mol]}$$

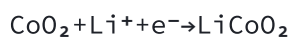
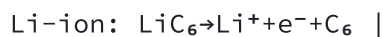
$$\alpha = \Lambda_m/\Lambda^{\circ}_m$$

$$\Lambda^{\circ}_m = \lambda^{\circ}_+ + \lambda^{\circ}_-$$

🔋 Battery Reactions



$$\text{Zn-Air } E_{\text{cell}} \approx 1.65 \text{ V}$$

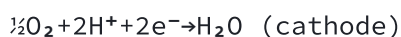
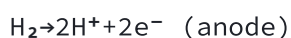


$$\text{Li-ion } E \approx 3.7 \text{ V nominal}$$

$$\text{Li-ion: } 150\text{--}250 \text{ Wh/kg}$$

$$\text{Zn-Air: } 350\text{--}500 \text{ Wh/kg}$$

🧴 Fuel Cell Reactions



$$E^{\circ}_{\text{cell}} = 1.23 \text{ V}$$

$$\Delta G^{\circ} = -237.4 \text{ kJ/mol}$$

PEM: Nafion® membrane

Only byproduct: H₂O

📊 Titration Key Points

Potentiometric: E vs $V \rightarrow$ inflection
= EP

Conductometric: G vs $V \rightarrow$ minimum =
EP (SA/SB)

Both: no colored indicator needed

First derivative max: EP location

HCl+NaOH curve: V-shaped

🚀 Sensor Comparison

Potentiometric $\rightarrow$ measures voltage

Amperometric $\rightarrow$ measures current

pH glass: 59.1 mV/pH unit

Ca²⁺ ISE: 29.6 mV/decade

Glucose sensor: linear i vs C

Clark electrode: O₂ measurement

$\text{CH}_3\text{COOH} + \text{NaOH}$: gradual then sharp $\uparrow$

Memory Tricks

OIL RIG: Oxidation Is Loss,
Reduction Is Gain

AN OX / RED CAT: Anode=Oxidation /
Cathode=Reduction

Higher E° = Cathode (always!)

Positive E°_{cell} = Spontaneous

H^+ fastest ion ($\rightarrow$ conductance $\downarrow$ when removed)

Li-ion = rocking chair battery

Standard E° Values

F_2/F^- : +2.87 V (strongest oxidant)

Cl_2/Cl^- : +1.36 V

$\text{O}_2/\text{H}_2\text{O}$: +1.23 V

Cu^{2+}/Cu : +0.34 V

H^+/H_2 (SHE): 0.00 V

Zn^{2+}/Zn : -0.76 V

Li^+/Li : -3.04 V (strongest reductant)

Top 10 Exam Tips:

1. Always identify cathode (higher E°) and anode (lower E°) first before any calculation
2. Nernst equation: $Q = [\text{products}]/[\text{reactants}]$ — don't confuse with K_{eq}
3. Cell notation: Anode (LEFT) | electrolyte || electrolyte | Cathode (RIGHT)
4. Molar conductance formula: use concentration in mol/L and κ in S/cm for Δm in $\text{S cm}^2/\text{mol}$
5. In conductometric titration of $\text{SA} + \text{SB}$: minimum conductance = equivalence point
6. Fuel cell vs Battery: fuel cell is continuous, battery stores energy
7. Li-ion charging reverses all reactions ($\text{LiCoO}_2 \rightarrow \text{CoO}_2 + \text{Li}^+$)
8. Potentiometric sensor: Nernst equation gives log relationship, not linear
9. Amperometric sensor: current is DIRECTLY proportional to concentration
10. At equilibrium $E = 0$, not $E^\circ = 0$. $E^\circ \neq 0$ unless same electrodes (concentration cell $E^\circ = 0$)

Quick Comparison Tables

Primary vs Secondary Batteries		
Feature	Primary (Zinc-Air)	Secondary (Li-ion)
Rechargeable	No	Yes (500–1500 cycles)
Cell voltage	~1.4–1.65 V	~3.6–3.7 V
Energy density	350–500 Wh/kg	150–250 Wh/kg
Electrolyte	KOH (alkaline)	LiPF ₆ in organic solvent
Use case	Hearing aids, watches	EVs, phones, laptops

Ionic Conductances at Infinite Dilution (298 K)		
Ion	λ° (S cm ² /mol)	Note
H ⁺	350	Fastest — Grotthuss mechanism (proton hopping)
OH ⁻	198	Second fastest — proton hopping in water
K ⁺	73.5	Similar size to Cl ⁻
Cl ⁻	76.3	Similar size to K ⁺
Na ⁺	50.1	Smaller ion but larger hydration shell
Ca ²⁺	119.0	High due to charge (+2)

Most Frequently Asked Exam Topics:

- Nernst equation derivation and calculation (appears in every exam)
- EMF calculation from standard electrode potentials
- Lithium-ion battery — construction, working, reactions
- Hydrogen fuel cell / PEMFC — construction and reactions
- Potentiometric and conductometric titration curves — interpretation
- Faraday's laws numerical problems (mass, time, current)
- Kohlrausch's law and molar conductance calculations
- ISE sensors — Nernst response, pH measurement

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/home/claude/electrochemistry.html
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